

#ASLI ENGINEERING

Pipelining

SWIPE

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Pipelining

A typical Redis flow is to send request, wait for response. Once response is received then send another request.



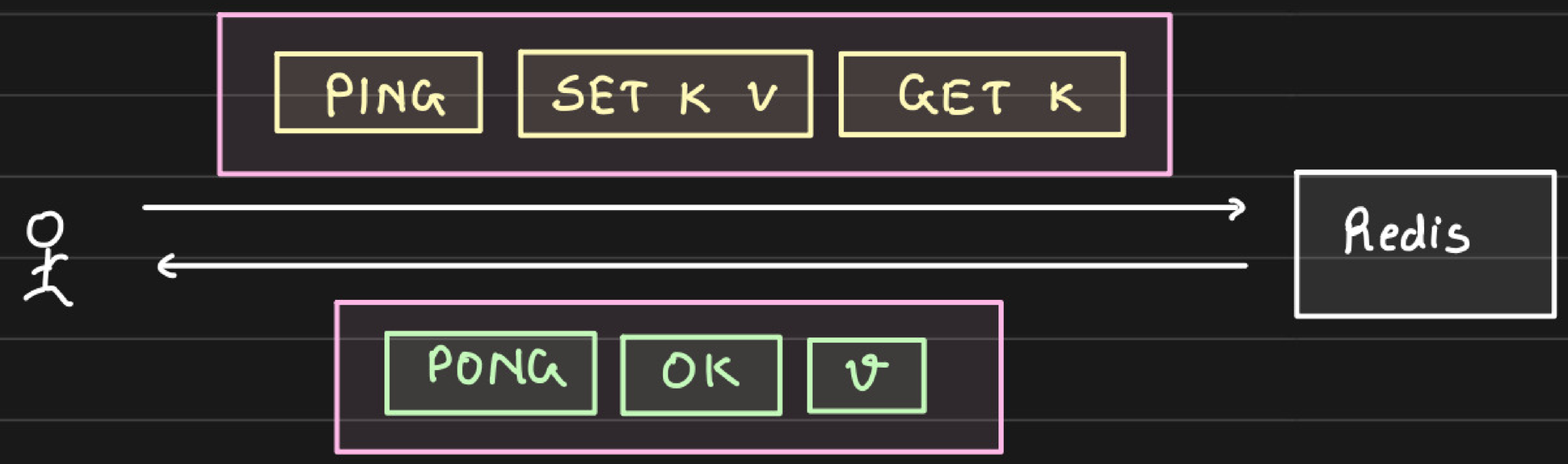
This classic flow has a major issue Round Trip Time

The RTT becomes significantly huge when there are large number of commands to be issued.

Hence, Redis provides Command Pipelining

Core Idea: Send multiple commands in one request

Receive output of each command in one response



Note: Pipelining ≠ Transaction

It is just about clubbing multiple command in one REQ and receiving individual outputs clubbed in a single RES

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With pipelining, client sends multiple commands to server
and server cannot respond until all are evaluated
Hence the output of each command is buffered until all are
evaluated and then server responds

↳ This would lead to a higher memory consumption

Improving Throughput

Pipelining not only improves on RTT but it also increases throughput

↓

commands per sec.

For Redis, evaluating a command

↳ in-mem operation

↳ superfast

But waiting, receiving, sending is I/O hence slower

With pipelining Redis server has to do fewer context switch to network IO
and focus more on executing commands.

↳ This substantially betters the throughput!

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